



Exhaust Emissions Testing of a Passenger Car with Compressed Natural Gas and Compressed Biogas Fuels for Assessment of Tank-to-Wheel Emissions

Moqtik Bawase^{1,2}, Ravi Sekhar^{1*}, Mohammad Jamadar², Yogesh Sathe², Sukrut Thipse²

¹ Symbiosis Institute of Technology (SIT), Pune Campus, Symbiosis International (Deemed University), 412115 Pune, India

² The Automotive Research Association of India (ARAI), 411038 Pune, India

* Correspondence: Ravi Sekhar (ravi.sekhar@sitpune.edu.in)

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Abstract: Compressed natural gas (CNG) is an alternative fuel that has less environmental impact than conventional fossil fuels. However, the availability of CNG is a constraint as it is a non-renewable source of energy and is being imported to meet the energy demand of India. Compressed Biogas (CBG), which is produced from renewable sources, has the potential to replace CNG. Due to renewable sources, the emissions from CBG are considered biogenic, and they do not contribute to the carbon bank of the atmosphere. However, the tailpipe emission quantification for automobiles can give an idea of the localised emission comparison for CBG and CNG. In this study, a detailed evaluation of tail-pipe emissions from the passenger car, with CNG and CBG fuels, was carried out using standard test protocol as per the Modified Indian Driving Cycle (MIDC). Statistical tools and techniques such as inter-fuel correlation, t-test, dynamic time warping (DTW), and Cosine Similarity test were utilised for critical evaluation of the emissions of the pollutants at different stages of the test cycle, like cold-phase, hot-phase, and extra-urban driving phase, to evaluate the emissions of CO, HC, NO_x, CO₂, and CH₄. Variability in emissions of CO, THC, NO_x, and CH₄ was observed in the cold-phase for CNG and CBG fuels. Aggregate CO, THC, and CH₄ tail-pipe emissions (mg/km) were found to be lower in the case of CBG than with CNG. Aggregate NO_x (mg/km) and CO₂ (g/km) emissions were higher in CBG. Significant variation in THC and CH₄ emissions was observed. CO₂ emissions were found to be similar for both fuels in all three phases. A marginal reduction (2%) in fuel efficiency with CBG compared to CNG was observed. Tank-to-wheel (TTW) greenhouse gas emissions of a passenger car with CNG as fuel were found to be about 24% lesser with CBG. The granular information generated in this study through a critical evaluation will be useful for engine designers for devising mitigation strategies to control the pollutant levels and to reduce their impact associated with air pollution exposure at ground level.

Keywords: Alternate fuel; Compressed bio gas; Compressed biogas; Compressed natural gas; Exhaust emissions; Life cycle analysis

1 Introduction

As per the International Monetary Fund 2025 estimates, India has surpassed Japan and become the 4th largest economy in the world. However, the economy is highly import-dependent for its energy requirements. According to the International Energy Agency (IEA) report, India has imported around 89% of crude oil and 47% of natural gas [1]. In the last ten years, the compound annual growth rate for imports of natural gas in India was 6%, as reported by the Ministry of Statistics and Program Implementation (MoSPI), Government of India [2].

Compressed natural gas (CNG) is an alternative fuel that has less environmental impact in comparison to conventional fossil fuels, i.e., petrol or diesel, as indicated in several studies [3–5]. Kuhnová et. al. [6] found that the greenhouse gas (GHG) emissions from CNG vehicles were reduced by 28%, NO_x emissions by 65%, and particulate matter emissions by 90% compared to diesel vehicles over a period of 5 years from 2015 to 2020. Although the emission benefits promote the replacement of fossil fuels by CNG, the availability of CNG in India is a constraint.

At present, the share of natural gas in India's energy mix is 6–7%, where the global average stands at 23.5% [7]. The maximum share of the natural gas consumption in India is by the fertiliser sector (31%), followed by the city gas distribution network, including transport, which utilised 20% of the total consumption of natural gas in India [8, 9]. It was found that around 19% of final energy was consumed in the transport sector, and two and four-wheeler passenger transport is taking 58% of the total share of the energy consumption in the passenger transport segment [10]. So, the alternative to CNG for passenger cars was explored, and compressed biogas (CBG) was found as the best alternative. The calorific value of CBG is similar to that of CNG, and the automobiles can run on this alternative fuel [11].

CBG is a purified form of biogas where the concentration of methane is enhanced, so that it can be used in automobiles. At first, biogas is generated from the anaerobic digestion (AD) of different feedstocks like cow dung, manure, swine, crop residues, municipal solid waste, organic, and food waste [12]. The generated biogas from the AD process contains around 60% of methane with impurities in the form of carbon dioxide, hydrogen sulfide, moisture, and trace gases. The biogas is purified via water scrubbing, chemical scrubbing, pressure swing adsorption, or membrane separation and compressed further to make it compatible to be used in automobiles [13]. With the extensive promotion of CBG by the government of India through the sustainable alternative to affordable transport (SATAT) scheme, with financial assistance to establish CBG plants by entrepreneurs and selling the manufactured CBG to oil marketing companies (OMCs) to be used in automobiles and industries, it is important to study the effects of CBG in detail.

The spark ignition engine performance was analysed with CBG as well as CNG as alternative fuels, and CBG showed better thermal efficiency and lower emissions of NO_x and hydrocarbons than CNG [14]. For a passenger car, a total of 10% reduction in nitrogen oxides and 36% reduction in non-methane hydrocarbons was realised with CBG. At the same time, the emission of carbon monoxide is lower in CNG than in CBG [15].

Since there are different purification methods implemented to produce CBG from biogas, the composition of CBG is quite variable. The variation in CO_2 is around 6% for different compositions of CNG (within the limiting range of IS 15958), and this variation gets doubled for CBG [16]. Understanding the emission levels of pollutants is important before putting any alternate fuel to use. This also works as an indicator of the combustion efficiency of the fuel in the SI engine.

Though the effect of biogas-derived fuels has been studied, there is limited research done to assess the tailpipe emission comparison between CNG and CBG in the Indian context for standard driving cycle conditions. In this study, a detailed evaluation of tail-pipe emissions from the passenger car using Modified Indian Driving Cycle (MIDC), a standard test protocol with CNG and CBG fuels, was carried out. A critical analysis was conducted to quantify and compare emissions of CO, HC, NO_x , CO_2 , and CH_4 . The emission behaviour in cold start, hot start, and extra urban driving cycle was also analysed to understand the combustion efficiency of the engine at various stages of test cycle. The emission data generated for CNG and CBG fuels will provide information on the overall emissions from the use phase of vehicles in the Indian context. Also, a very detailed analysis will enable engineers to decide the strategies for improving the combustion efficiency and performance of aftertreatment devices for the reduction in tailpipe emissions.

2 Methodology

Considering the similar characteristics of fuels, CNG and CBG emissions levels are expected to be similar. To objectively evaluate the emissions from both of these fuels, standard emission tests are performed on a passenger car, with both fuels using the MIDC emission cycle. Emission cycles are emission variations with a sequence of different speeds and load conditions performed on an engine or chassis dynamometer. The final test results are obtained by analyzing exhaust gas samples collected in polyurethane bags over the duration of the cycle. Repeatability of the test was established for each fuel by repeating the test twice with each fuel. To understand the variation, if any, in the emission levels with both the fuels, results are evaluated for cold, hot, and extra urban driving phases with phase-I representing cold start (0–195 s), phase-II representing hot conditions (196–780 s), and phase-III representing extra urban driving cycle (781–1,180 s). Outcomes are statistically evaluated for identifying the significance of the difference (t-test, phase, directional) in observations.

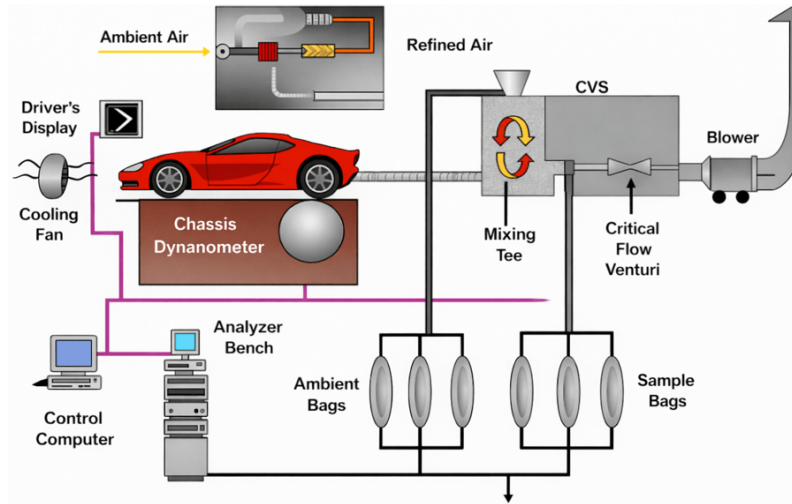
2.1 Test Setup

Emissions measured on chassis dynamometers are expressed in grams of pollutant per unit of travelled distance, expressed in g/km. In a transient cycle such as the one used for the chassis dynamometer testing, the vehicle follows a prescribed driving pattern, which includes accelerations, decelerations, changes of speed, and load. Specifications of the test set-up are presented in Table 1.

A 150-kW chassis dynamometer having 48 inches roller diameter with an inertia simulation range from 454 to 5448 kg was used for complete road load simulation, measurement of vehicle tail-pipe emissions. The latest generation dilute exhaust gas analyzer, constant volume sampler (CVS), and Dilution tunnel suitable for measurement of four-wheeler passenger cars was used. A schematic layout of the test cell with the vehicle is given in Figure 1.

Table 1. Specifications of the test set-up

Parameter	Specification
Chassis Dynamometer Specifications	
Maximum Permissible Axle Load (kg)	4500
Roller Diameter (mm)	1219.2
Distance Between Roller Outer Edges (mm)	2744
Distance Between Roller Inner Edges (mm)	914
Base Inertia (kg)	1209
Inertia Simulation Range (kg)	454–5448
Minimum Permanent Motoring Power (kW)	150
Speed Range (km/h)	up to 200
Emission System Specifications	
Exhaust Gas Sampling System	Constant Volume Sampler-Critical Flow Venturi (CVS-CFV)
CVS Flow (m^3/min)	4–30
Venturi Sizes	4, 6, 8, 12
Heated Bag Cabinet	12 Bags (4 + 4 + 4)
CO Analyzer Range (ppm)	0–5000
CO ₂ Analyzer Range (%)	0–6
THC, CH ₄ , NO _x , NO (ppm)	0–1000

**Figure 1.** Schematic diagram of chassis dynamometer testing cell

2.2 Test Vehicle

A BS-VI compliant Passenger Car was used for conducting the tests. The key specifications of the test vehicle are given in Table 2. The engine and other vehicle systems were not modified to compare the emission performance of both fuels.

2.3 Test Fuels

The test fuels, CNG and CBG, meeting the fuel quality specifications were sourced from the retail outlets. The CBG used in the study was produced from agricultural residue (rice husk) through AD and subsequent purification using pressure swing adsorption. Composition of the CNG and CBG fuels used with respect to the fuel specifications IS 15958 [17] for CNG and IS 16087 [18] for CBG are given in Table 3 and Table 4 respectively.

2.4 Test Protocol

Emission test on the vehicle was performed in an emission test cell by following a standard MIDC [19]. MIDC has two phases (Figure 2). The first phase is an urban driving phase having a maximum speed of 50 kmph. The second phase is the highway phase (Extra Urban Driving Cycle), having a maximum speed of 90 kmph. The cycle has a total duration of 1,180 s, further divided into two parts: urban driving cycle (780 s) and extra urban driving cycle (400 s), and covers 10.647 km.

Table 2. Key specifications of test vehicle

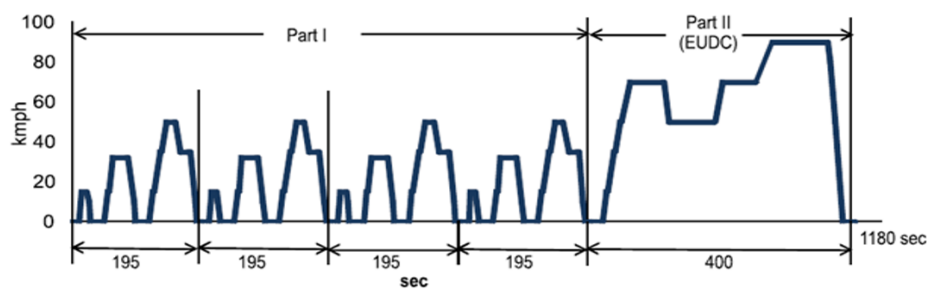
Parameter	Specification
Type	BS-VI Passenger Car
Engine Displacement (cc)	1462
Inertia Class (kg)	1360
Kerb Weight (kg)	1235
Gross Vehicle Weight (kg)	1695
Rated Power	66 kW @ 5500 rpm
Maximum Torque	122 Nm @ 4200 rpm
Cooling System	Water cooling
Fuel Type	Compressed Natural Gas (CNG)
Max Power	64.6 kW @ 5500 rpm
Max Torque	121.5 Nm @ 4200 rpm
Fuel Efficiency	26.6 km/kg
Fuel Tank Capacity	55 L (Water Equivalent)
Transmission Type	5MT
Drive Type	2WD
Wheel Size	215/60 R17

Table 3. Composition of compressed natural gas (CNG) fuel used for testing

Properties	CNG Requirements (IS 15958)	CNG Test Fuel (Measured)
Methane (% vol, min)	90.0	94.14
Ethane (% vol, max)	6.0	2.39
C3 + HC (% vol, max)	3.0	1.71
C6 + HC (% vol, max)	0.5	0.08
Total Sulphur (mg/m ³ , max)	10.0	8.80
Oxygen (% vol, max)	0.5	0.02
CO ₂ (% vol)	–	1.44
CO ₂ + N ₂ (% vol, max)	3.5	1.68
Water Content (mg/m ³ , max)	5.0	<5.0
Hydrogen (% mole, max)	0.1	<0.01
Carbon Monoxide (% mole, max)	0.1	<0.01

Table 4. Composition of compressed biogas (CBG) fuel used for testing

Properties	IS 16087 Specification	CBG Sample
CH ₄ (% min)	90	94.7
Moisture (mg/m ³ , max)	5	<5.0
H ₂ S (mg/m ³ , max)	20	3.0
CO ₂ + N ₂ + O ₂ (% max)	10	5.4
CO ₂ (% max)	4	3.1
O ₂ (% max)	0.5	0.1

**Figure 2.** Modified Indian driving cycle (MIDC)

Drive wheels of the test vehicle were mounted on the test-bed rollers. To simulate aerodynamic drag and friction, rotational resistance was adjusted. The weight of the vehicle was simulated by adding inertial mass. A variable-speed cooling blower was mounted in front of the vehicle to provide cooling. To bring the conditions of the test vehicle and the engine to the test cell ambient conditions, the test vehicle was soaked. The chassis dynamometer was warmed up before starting the test after mounting the vehicle and keeping it in engine-off condition. Subsequently, to compensate for the frictional losses, the chassis dynamometer was calibrated, and the test was commenced. The vehicle was driven on a chassis dynamometer by maintaining speed and changing gears. A dilution tunnel and a critical flow venturi-type CVS were used to dilute engine-out exhaust gases with fresh air. Diluted exhaust gas was extracted and collected in a bag by drawing a constant proportion for gaseous emission measurement. The analysis of the gaseous pollutants (CO, THC, NO_x, CO₂, and CH₄) was undertaken immediately after the test using the gas analysers, which were calibrated before the start of the test. Span gases were used for calibration for each pollutant, depending on the range of the pollutant analysers. The results of a test were considered to be acceptable when the difference in observed value is less than 2% span gas concentration or 0.3% of the full-scale reading of a pollutant analyser. The validation tests were performed to assess the drift between pre-test and post-test values, which should be less than 1% of the full-scale reading of the pollutant analyser. As the concentration of the gases in the sample bag was proportional to the mean exhaust gas-fresh air mixture, and the total volume of the exhaust-air mixture is known, the mass of pollutants produced during the test was calculated, and the emissions of pollutants in g/km were calculated using Eq. (1):

$$M_i = (V_{\text{mix}} * Q_i * k_H * C_i * 10^{-6}) / d \quad (1)$$

where,

M_i : mass emission of the i^{th} pollutant in g/km;

V_{mix} : volume of the diluted exhaust gas expressed in m³/test and corrected to standard conditions 293 K and 101.33 kPa;

Q_i : density of the i^{th} pollutant in kg/m³ at normal temperature and pressure (293 K and 101.33 kPa);

k_H : humidity correction factor, a constant used for the calculation of the mass emissions of oxides of nitrogen only;

C_i : concentration of the i^{th} pollutant in the diluted exhaust gas expressed in ppm and corrected by the amount of the i^{th} pollutant contained in the dilution air;

10^{-6} : is the conversion factor to change concentration unit ppm into a dimensionless fraction; and

d : distance travelled by the vehicle during the test in km.

2.5 Statistical Evaluation

To objectively identify differences or similarities in the test results between the pollutants with CNG and CBG fuels, a statistical evaluation was carried out for the results obtained in three distinct phases, namely cold, hot, and extra-urban phases.

Inter-fuel correlation for different pollutants was assessed to identify the correlation of the time series values obtained with two fuels for all three phases. A higher correlation of a pollutant between two fuels will indicate similar emission behaviour.

To identify whether the emissions with CNG and CBG are significantly different, a t-test is performed, which compares the means of pollutants in the phases assuming normal distribution. P -values were determined to identify the difference, if any, in the mean values.

Further, dynamic time warping (DTW) and Cosine Similarity analysis of the emission data with CNG and CBG fuels were carried out to quantify the difference in emissions. DTW aligns time series that may be out of phase with respect to time. If the peaks of pollutants are attained at different time intervals with two fuels, higher DTW values are obtained. This will indicate that more temporal alignment of time series is required. Cosine Similarity measures the directional similarity of pollutant vectors with two fuels. Cosine similarity values are closer to one if a pollutant is following a similar directional pattern with both the fuels.

3 Results and Discussion

3.1 Repeatability of Testing

Aggregated mass emission test results of the complete MIDC with CNG and CBG fuels are presented in Table 5. Two tests (T1: Test1 and T2: Test2) were performed using the standard test protocol with each fuel. Mean and standard deviation (SD) of both the test results are presented for all the pollutants. Emissions of all the pollutants for repeat testing, with the same fuel, were found to match closely with each other.

The aggregate emissions over the entire driving cycle for CO₂ (g/km) show higher emissions with CBG than with CNG. This can be attributed to the presence of CO₂ gas in the CBG fuel (3.1%), as can be seen in Table 3, which is

adding to the overall emissions from the exhaust. About 3.4 g/km (out of 111.1 g/km) of CO₂ can be attributed to the CO₂ in the fuel. Carbon monoxide (CO) and Hydrocarbons (THC), which are products of incomplete combustion, were found to be lower in CBG (189.4 and 48.6 mg/km for CO and TH respectively) than in CNG (205.5 and 65.9 mg/km for CO and THC, respectively), indicating better combustion efficiency. The higher Oxides of Nitrogen (NO_x) emissions may be due to the high temperature thermal NO_x formation, due to efficient combustion, in CBG. Emissions of methane (CH₄) were observed to be slightly more than half of THC (53%) in the case of CNG and CBG. Emission levels of the pollutants like CO, THC, NO_x, and non-methane Hydrocarbons (NMHC) were found to be well below the Bharat Stage-VI (BS-VI) emission limits of 1,000 mg/km, 100 mg/km, 60 mg/km, and 68 mg/km, respectively, in the case of both the fuels [20]. Calculated fuel efficiency, which is based on the carbon balance method, was higher with CNG (16.4 km/m³) than with CBG (16.1 km/m³). This reduction in fuel efficiency, of about 1.8%, can be attributed to the higher concentration of CO₂ in CBG. A detailed analysis of the aggregate levels of pollutants in the exhaust with CNG and CBG is carried out in the following section through the variations observed in the time series with the MIDC.

Table 5. Results of mass emission tests (T1: Test1 and T2: Test2) with compressed natural gas (CNG) and compressed biogas (CBG) fuels

Fuel/Test	CO ₂ (g/km)	CO (mg/km)	THC (mg/km)	NO _x (mg/km)	CH ₄ (mg/km)	Fuel Efficiency (FE, km/m ³)
CNG T1	109.0	203.2	68.7	10.8	36.2	16.4
CNG T2	109.3	207.7	63.0	11.9	33.8	16.3
CNG Mean	109.2	205.5	65.9	11.4	35.0	16.4
CNG SD	0.2	3.2	4.0	0.8	1.7	0.0
CBG T1	111.2	183.5	49.3	14.3	25.9	16.1
CBG T2	110.9	195.3	47.8	18.1	25.4	16.1
CBG Mean	111.1	189.4	48.6	16.2	25.7	16.1
CBG SD	0.2	8.4	1.1	2.7	0.3	0.0

3.2 Temporal Variation in Emissions

Mass emission test results of the vehicle with CNG and CBG fuels are presented in the form of a time series in Figure 3. Test vehicle speed (Figure 3a) and acceleration (Figure 3b) are plotted over the entire time span of 1,180 s of MIDC as a time series for CNG (red) and CBG (blue) with a frequency of 1 Hz. Time series of speed (kmph) and acceleration (m/s²) can be seen overlapping each other, indicating that the MIDC test cycle for CNG and CBG was practically followed during the test, and the outcome of the emissions for different pollutants can be compared. A comparison of various species, such as CO₂, CO, THC, NO_x, and CH₄ (all in ppm) during the driving cycle is presented. CO emissions were found to be higher in the case of CNG than CBG in the cold start phase, i.e., up to 195 s (Figure 3c). However, CO emissions remained somewhat similar during the hot cycle phase (196–780 s) and extra-urban driving phase (781–1,180 s). In the case of THC, higher values were observed for CNG in the cold phase as well as the extra-urban driving cycle (Figure 3d). Lower CO and THC emissions in CBG indicate better combustion efficiency than CNG. This is reflected in higher NO_x emissions with CBG in the cold phase due to the formation of thermal NO_x at higher combustion temperature (Figure 3e), which remained similar for the rest of the test cycles. Higher aggregate CH₄ emissions with CNG can be mainly attributable to higher methane slip at higher speeds encountered during the extra-urban driving cycle (Figure 3g), which was reflected in higher total hydrocarbon emissions also (Figure 3d). In addition, a higher peak of CH₄ was observed in CNG during the cold phase, which is attributable to incomplete combustion. Slightly higher emission levels of CO₂ in the case of CBG than CNG throughout the test cycle are observed (Figure 3f). This is in line with the results of overall high aggregate CO₂ levels in CBG due to the presence of higher CO₂ in CBG fuel.

3.3 Variation in Emissions with Speed and Acceleration

A density scatter plot of pollutant concentration (ppm) with vehicle speed, ranging from 0 to 90 kmph, and acceleration is presented in Figure 4. The colour of the data points represents data density, from blue for a few data points to red for higher data density, for a particular concentration of pollutants. The first and second columns of Figure 4 are for the density scatter plot of speed and pollutant concentration with CNG and CBG respectively, while the third and fourth columns are for the density scatter plot of acceleration and pollutant concentration with CNG and CBG respectively. Most data points for CO, THC, NO_x, and CH₄ are concentrated at lower vehicle speeds, especially between 0 and 40 kmph and lower concentrations with CNG (Figures 4(A1), 4(B1), 4(C1), and 4(E1)) and CBG (Figures 4(A2), 4(B2), 4(C2), and 4(E2)). A few higher concentration points are observed at higher speeds for THC and CH₄. However, in the case of CO₂, though the highest density of data points was at low speed and low concentration, a spread of the data was observed at all the speeds for both the fuels (Figures 4(D1) and 4(D2)).

Figure 4 shows that the maximum number of data points is at a constant speed or at zero acceleration. The highest data density was found to be near zero acceleration and low concentrations for CO, THC, NO_x, and CH₄ for both CNG (Figures 4(A3), 4(B3), 4(C3), and 4(E3)) and CBG (Figures 4(A4), 4(B4), 4(C4), and 4(E4)). Though for CO₂, the spread of the concentration was higher, ranging from very low to 20,000 ppm, the density of the data points was maximum around the zero acceleration (or constant speed). The data density was somewhat higher on the acceleration side than in the deceleration side for both CNG (Figure 4(D3)) and CBG (Figure 4(D4)).

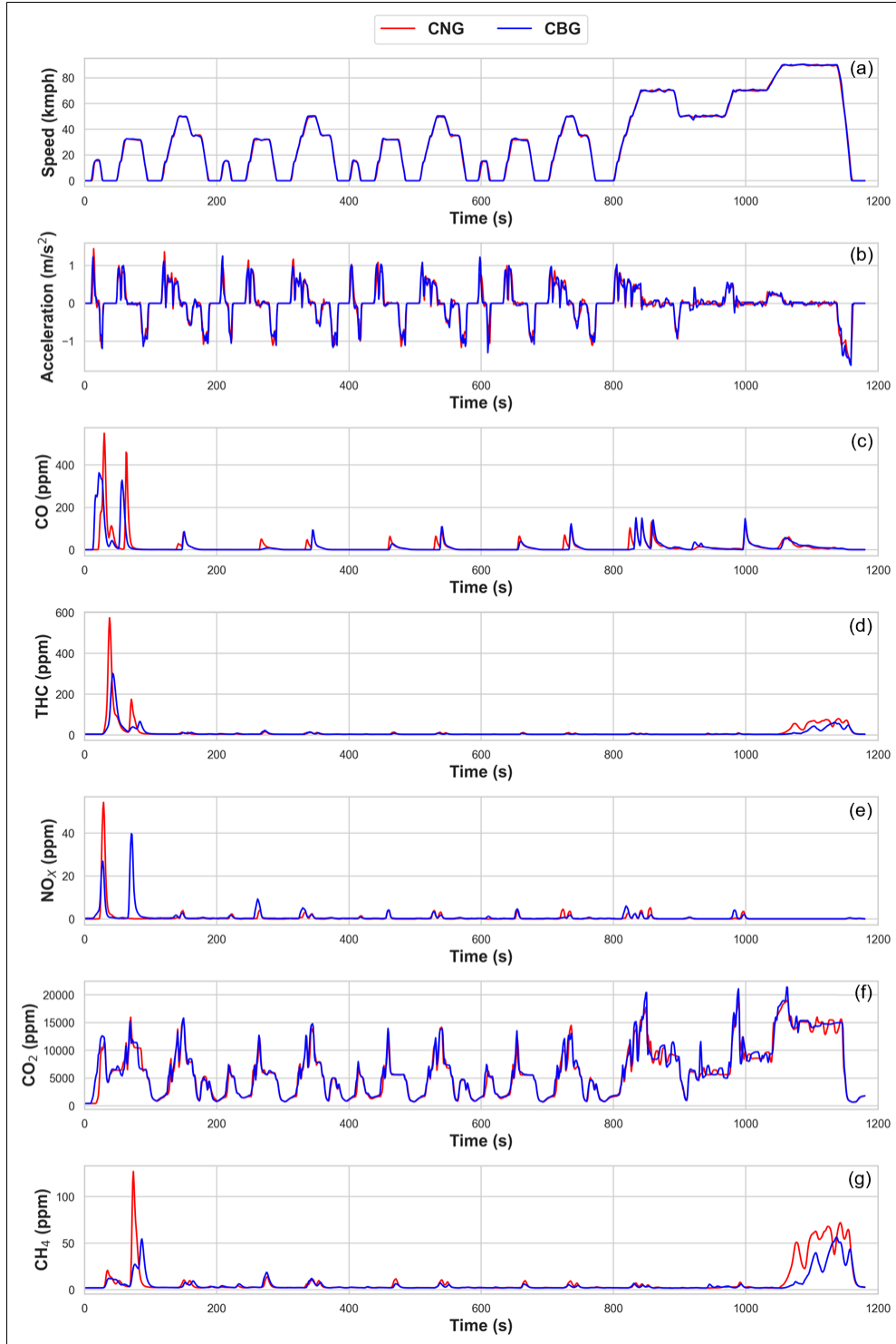


Figure 3. Comparison of the time series of test cycle and emissions for compressed natural gas (CNG) and compressed biogas (CBG) testing

Note: a-Speed, b-Acceleration, c-CO, d-THC, e-NO_x, f-CO₂ and g-CH₄

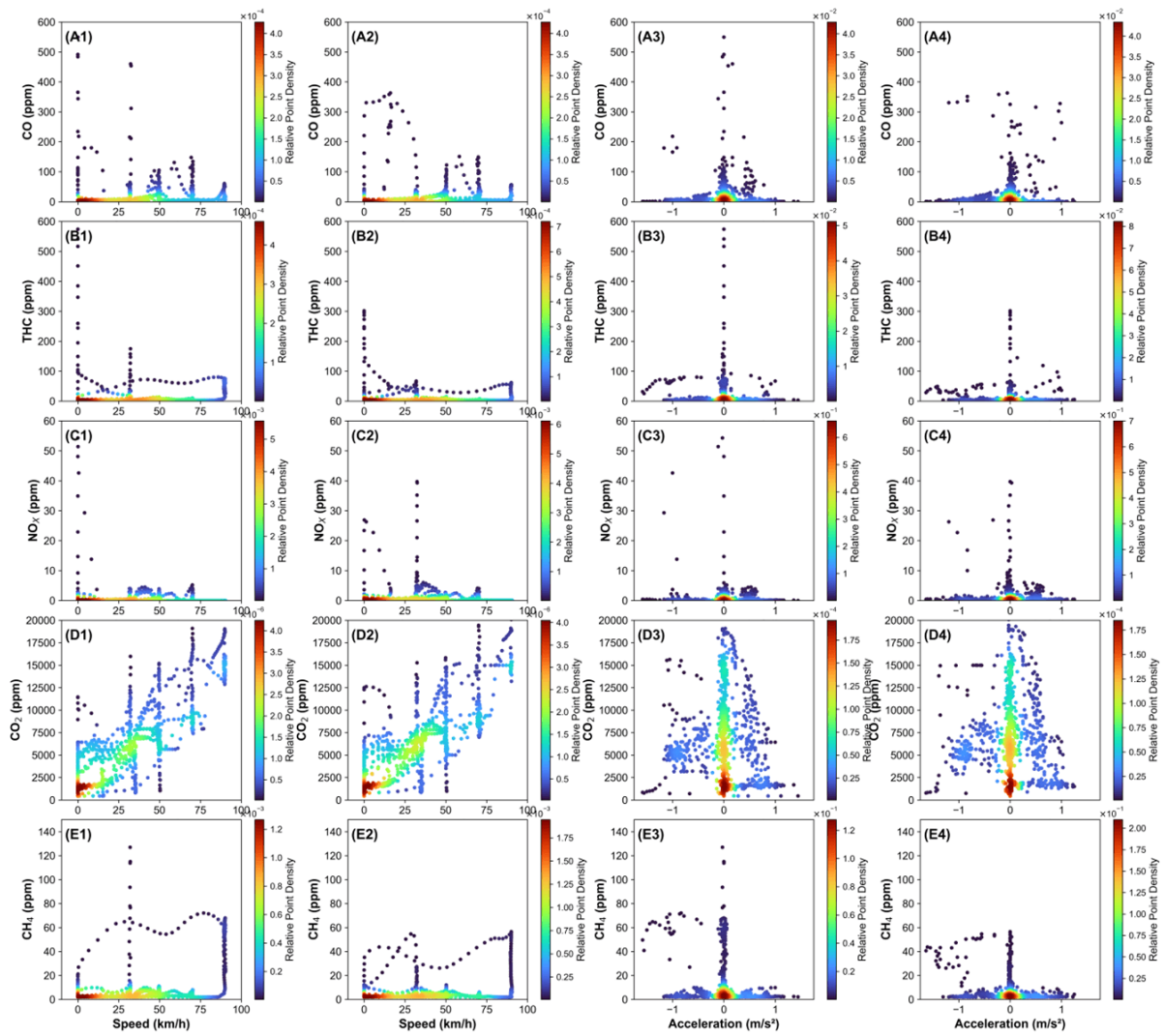


Figure 4. Density plots of speed, acceleration, and pollutants

Note: A-CO, B-THC, C-NO_x, D-CO₂, and E-CH₄; 1: Compressed natural gas (CNG)-Speed; 2: Compressed biogas (CBG)-Speed; 3: CNG-Acceleration; 4: CBG-Acceleration

It was observed from Figure 4 that emissions of all the pollutants, excluding CO₂, were occurring at lower speeds and around zero or at very low acceleration/deceleration levels. CO₂ emissions were found to spread across the speed range and also at different acceleration/deceleration levels. To assess the emissions of various pollutants at different speeds and acceleration combinations, 3D scatter plots of pollutants against speed and acceleration for CNG and CBG are presented in Figure 5.

Analysis of the 3D scatter plots (Figure 5) shows that acceleration/ deceleration at the X-axis ranges from -1.4 to 1.0 m/s², speed at the Y-axis ranging from 0 to 90 kmph and emission levels of pollutants (a-CO, b-THC, c-NO_x, d-CO₂ and e-CH₄) are presented at the Z-axis.

A large number of data points of CO emissions are found to be concentrated at zero acceleration, meaning constant speed plateaus (at 32, 50, 70, and 90 kmph) of the driving cycle (Figure 5a). Maximum concentrations of CO were also found at around zero acceleration. A larger number of higher CO emission points were found to fall in the deceleration mode during lower speeds (up to 20 kmph). This higher concentration of CO at constant speeds and deceleration mode can be attributed to the lower combustion temperatures, leading to incomplete combustion. The density of THC emissions points was highest at zero acceleration. Also, the points of higher concentrations, up to 500 ppm for CNG and 350 ppm for CBG, were also found at zero acceleration and low speed (Figure 5b). Mid-level concentrations of range from 50 to 100 ppm were largely in the deceleration mode across the entire speed range. Though overall concentrations were lower, a similar pattern was observed for CH₄ also (Figure 5e). However, a significant number of points were also falling in the acceleration mode at lower speeds in the case of THC, which was not that evident in the case of CH₄. Maximum NO_x concentrations were found to occur at zero acceleration

(Figure 5c). Some higher NO_x concentration data points were found to be in deceleration mode at lower speeds. During acceleration mode, most of the data points were at low concentration across all speeds. The density of data points for CO_2 concentrations was high at zero acceleration (Figure 5d). However, CO_2 data points were spread across all the acceleration and deceleration ranges, with higher data points in acceleration mode than in deceleration mode. CO_2 concentrations at higher speeds were found to be higher than 7,500 ppm across all acceleration and deceleration levels, as can be seen in Figures 4(D1) and 4(D2).

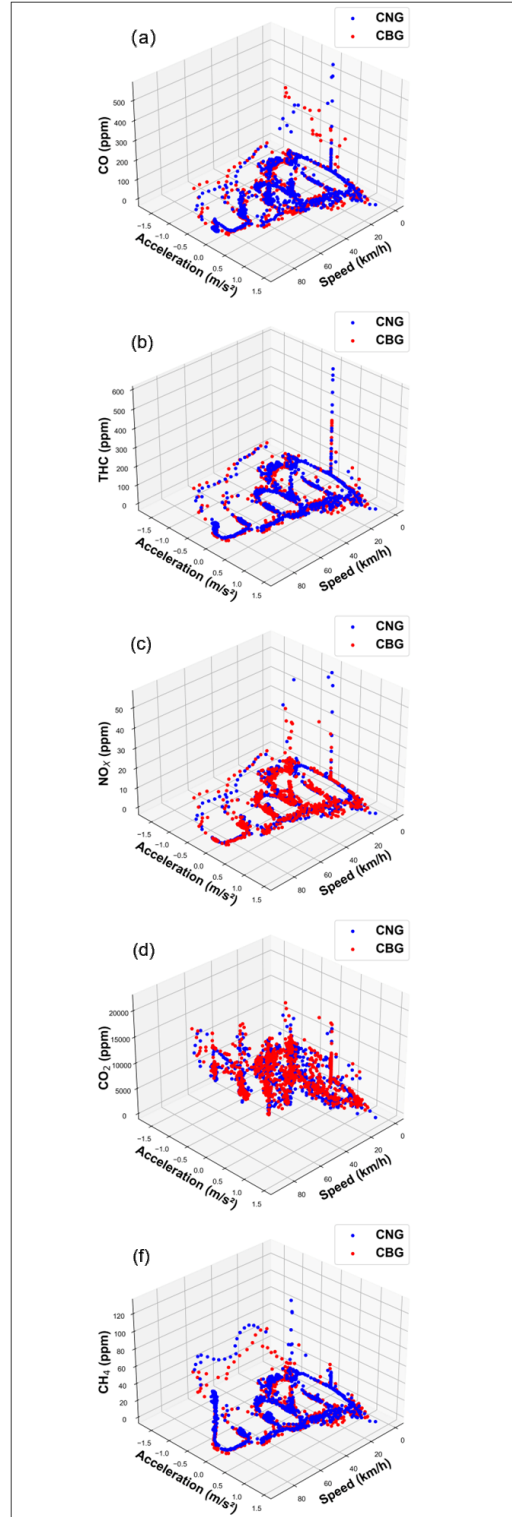


Figure 5. 3D scatter plots of various pollutants in ppm

Note: a-CO, b-THC, c- NO_x , d- CO_2 and e- CH_4 against speed (km/h) and acceleration (m/s^2) for compressed natural gas (CNG, blue) and compressed biogas (CBG, red)

3.4 Statistical Evaluation of Pollutants with Compressed Natural Gas (CNG) and Compressed Biogas (CBG)

Statistical assessment was carried out for the test results between the pollutants with CNG and CBG fuels (Table 6) in three distinct phases of the MIDC test cycle, such as cold start (0–195 s), hot conditions (196–780 s), and extra-urban driving cycle (781–1,180 s).

Table 6. Inter-fuel correlation and t-test of pollutant emissions with compressed natural gas (CNG) and compressed biogas (CBG) fuels

Pollutant	Phase	Inter-Fuel Correlation	T-Test <i>P</i> -Value
CO ₂	P-I	0.93	0.47
	P-II	0.97	0.31
	P-III	0.97	0.54
CO	P-I	0.34	0.74
	P-II	0.79	0.01
	P-III	0.85	0.93
THC	P-I	0.58	0.22
	P-II	0.80	0.80
	P-III	0.87	<0.05
NO _x	P-I	0.49	0.32
	P-II	0.65	<0.05
	P-III	0.59	0.02
CH ₄	P-I	0.45	0.31
	P-II	0.81	0.35
	P-III	0.88	<0.05

The correlation of mean values obtained for all three phases for CO₂ was found to be in very good agreement (above 0.9). Correlation for CO, THC, NO_x, and CH₄ in phase-I was low, indicating a difference in emission levels in the cold phase (below 0.6). The lowest correlation was observed in the case of CO emission in phase-I. CO, THC, and CH₄ emissions were very well correlated (more than 0.75) in phase-II and phase-III. NO_x emissions were not correlated (values ranging from 0.49 to 0.65).

The *p*-values obtained from the t-test for different pollutants with CNG and CBG are shown in Table 6. *P*-values less than 0.05 indicate a significant difference between the emission of pollutants with CNG and CBG. It can be seen that CO and NO_x emissions in phase II are significantly different. Similarly, THC and CH₄ emissions during phase-II are significantly different in CNG and CBG. This can also be seen in the time series graphs presented in Figures 3e and 3g.

Table 7. Statistical evaluation of pollutant emissions with compressed natural gas (CNG) and compressed biogas (CBG) fuels

Pollutant	Phase	Dynamic Time Warping (DTW) Distance (Norm.)	Cosine Similarity
CO ₂	P-I	1.00	0.98
	P-II	0.66	0.99
	P-III	0.00	0.99
CO	P-I	0.00	0.45
	P-II	0.83	0.82
	P-III	1.00	0.90
THC	P-I	0.00	0.65
	P-II	1.00	0.96
	P-III	0.87	0.92
NO _x	P-I	0.00	0.53
	P-II	0.90	0.72
	P-III	1.00	0.65
CH ₄	P-I	0.00	0.59
	P-II	1.00	0.94
	P-III	0.52	0.91

An additional statistical analysis of the emission data with CNG and CBG fuels was carried out to quantify the

difference in emissions in terms of DTW, which aligns time series that may be out of phase with respect to time, and Cosine Similarity to measure directional similarity of pollutant vectors. The results for different pollutants are presented in Table 7 for three different phases as mentioned above. Values obtained for DTW distance are normalised to present the relative differences among the values in the segment for a pollutant. Value of 1 against a phase denotes the highest similarity in the level of a pollutant in that phase. Values closer to 1 show more similarity, and a value of zero against a phase represents the least similarity among the phases. In the case of cosine similarity, a value close to 1 indicates more similarity in directional vectors.

Normalised DTW distance of THC and CH₄ was lowest (normalised to 1) in phase-II, indicating the highest similarity or temporal alignment, which was followed by phase-III. In the case of CO and NO_x, temporal alignment was highest in phase-III and was closely followed by phase-II. However, in CO₂, the highest temporal alignment was observed in phase-I followed by phase-II and the lowest in phase-III. Lowest temporal alignment was observed for CH₄ in phase-I and phase-III, which can be seen in Figure 3g. The analysis revealed that temporal variability for CO, THC, NO_x, and CH₄ was highest in phase-I, which is the cold phase, and can be attributed to incomplete combustion. Evaluation of Cosine Similarity of the emissions of pollutants with CNG and CBG suggests that CO₂ has the highest directional similarity in all phases. That means CO₂ is rising and falling together with both the fuels. CO, THC, and CH₄ showed significant directional similarity in phase-II and phase-III. It is to be noted that though there are significant changes in the magnitude of pollutants such as THC and CH₄ in phase-III, still due to movement of the emissions in the same direction, increasing or decreasing at the same time in both fuels, the directional similarity is high. However, in phase-I it was found to be significantly low for CO, THC, and CH₄. Directional similarity of lowest for NO_x in all the phases.

3.5 Tank-to-Wheel (TTW) Emissions

CO₂ and CH₄ are the greenhouse gases that are emitted from the tailpipe of a vehicle. From the tailpipe emission data (g/km) for CO₂ and CH₄, the TTW emission from the entire life cycle of a vehicle can be calculated. In this study, tail-pipe emissions of passenger cars for CO₂ were found to be 109.2 and 111.1 g/km respectively, for CNG and CBG, and CH₄ emissions of 35.0 and 25.7 g/km for CNG and CBG respectively. Considering the total travel km of 150,000 in an entire lifespan of a passenger car vehicle, the TTW emission can be calculated in terms of tonnes of CO₂ equivalent using global warming potential values of CO₂ and CH₄ reported under the Greenhouse Gas Protocol [21]. TTW greenhouse gas emissions of a passenger car with CNG as fuel were estimated as 172,800 kgCO₂eq. And with CBG, it was found to be 131,500 kgCO₂eq. meaning about 24% lesser GHG emissions with CBG.

It is to be noted that due to the biogenic origin of the biogas, it is considered carbon neutral, and the greenhouse gas emissions are considered to be zero under international standards for accounting, like the GHG protocol. However, assessment of the tailpipe emissions is important for devising mitigation strategies to control the pollutant levels to reduce their impact associated with air pollution exposure at the ground level. Also, emission testing provides necessary inputs like fuel efficiency with alternate fuels like CBG, and thus the total amount of fuel required over its usage. This information is an essential input for conducting life cycle assessment, particularly of the production and distribution stages.

4 Future Research Scope

In addition to the emission testing with pure CNG and CBG gases, studies on different blends of CBG and CNG can also be conducted to understand the impacts on exhaust emissions. The results from this paper will help in establishing the baseline for studies with higher quality of CBG. Also, the overall life cycle assessment for identifying environmental impacts, including production, its use in the vehicle till end-of-life, will provide more insights into the environmental impacts associated and identify critical stages in its life cycle. The manufacturing of biogas is a mature technology; however, the purification techniques need to be studied in detail, and life cycle analysis (LCA) analysis will help in identifying the hotspots in the process to have efficient conversion of biogas to CBG. This will enable addressing the challenges for effective implementation of the government scheme to promote CBG and will motivate researchers to work on the solutions of the critical hotspots.

5 Summary and Conclusions

The present work studied the exhaust emissions of different regulated pollutants with CNG as a fossil fuel source and CBG produced from agricultural residue (rice husk) through AD and subsequent purification. Gaseous fuel samples were random and were obtained from the retail outlets. Emission tests were performed using the standard test method on a BS-VI compliant vehicle with CNG and CBG fuels. The tests were repeated twice, each with CNG and CBG.

Aggregate CO, THC, and CH₄ tail-pipe emissions (mg/km) were found to be lower in the case of CBG than CNG when tested as per MIDC. Aggregate NO_x (mg/km) and CO₂ (g/km) emissions were higher in CBG. Emission

data points were analysed for frequency distribution with respect to speed, acceleration/deceleration to understand the combustion phenomenon. Highest density of emissions and peak emissions data points for CO, THC, and NO_x were found at low speeds and zero acceleration levels. In the case of CH₄, peak values of emissions were observed at high speed and zero acceleration, but the density of data points was high at low speeds. CO₂ emissions were found to increase with speed and are spread across all acceleration levels.

Emission data, with 1 Hz frequency, obtained through the MIDC test cycle, were further categorised into three distinct phases: cold start (phase-I from 0–195 s), hot-phase (phase-II from 196 to 780 s), and extra-urban cycle with high speed up to 90 kmph (phase-III from 781–1,180 s) for further analysis of emission behaviour of different pollutants. Statistical assessment of the emission data was performed on the data observed for different pollutants in these three phases.

Interfuel correlation coefficient of CO, THC, NO_x, and CH₄ in phase-I was comparatively lower, indicating inconsistent combustion/emission patterns. However, it improved significantly in phase-II and phase-III, suggesting that more consistent emission trends in CNG and CBG may be due to more stable or optimized engine conditions. In CO₂ emissions, a very good correlation between CNG and CBG was observed in all three phases.

The tail-pipe emission data, when subjected to independent sample T-tests to evaluate the statistical significance of differences in emissions between CNG and CBG fuels across three testing phases, revealed that p-values observed at 95% confidence interval were <0.05 for THC and CH₄ in phase-III, suggesting that the fuel type has a measurable impact on the emission characteristics of unburned hydrocarbons and methane under Phase-III conditions. In the case of CO, a significant difference was observed in phase-II, and in NO_x emissions, it was observed in phase-II and phase-III, suggesting differential nitrogen oxide formation mechanisms or combustion dynamics between CNG and CBG. No significant difference was observed in CO₂ emissions in all three phases with CNG and CBG fuels. The observed difference in the emission levels was further verified by applying DTW and Cosine Similarity tests to assess temporal alignment and directional similarity respectively. Results of DTW distance, Cosine similarity, and Inter-fuel correlation are corroborated very well and indicate variability in emissions of CO, THC, NO_x, and CH₄ in the phase-I for CNG and CBG fuels. CO₂ emissions were found to be similar for both fuels in all three phases. The granular information generated in this study through a critical evaluation may be used by engine designers for further improvement in the performance and tuning of the engine for better combustion efficiency.

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Author Contributions

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Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

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Nomenclature

$Q_{(i)}$	Density of pollutant, kg/m ⁻³
$C_{(i)}$	Concentration of pollutant, ppm
$k_{(H)}$	Dimensionless humidity correction factor
kgCO ₂ eq.	Greenhouse gas emissions in carbon dioxide equivalent, kg
i	Pollutant particle